Experiment: Design and Implement an Enterprise Architecture Framework

1. Title

Design and Implementation of an Enterprise Architecture Framework for a Hypothetical Organization using EdrawMax

2. Problem Statement

Design and develop an **Enterprise Architecture (EA) framework** for a hypothetical organization that integrates its **business architecture, application architecture, data architecture, and technology architecture**.

The architecture should represent how business objectives and processes are supported by applications, data, and underlying technology infrastructure. The proposed architecture should provide a structured view of the organization and help in effective planning, integration, decision-making, security, and future technology development.

The enterprise architecture will be designed and represented using **EdrawMax**.

3. Software Requirements

Software Used

EdrawMax

EdrawMax is a diagramming and visualization software that can be used to create:

- Enterprise architecture diagrams
- Business architecture diagrams
- System architecture diagrams
- Network diagrams
- Flowcharts
- Organizational charts
- UML diagrams
- Data flow diagrams

Hardware Requirements

- Computer/Laptop
- Minimum 4 GB RAM
- Minimum 2 GB free storage
- Keyboard and mouse
- Internet connection, if required for online version

Software Requirements

Requirement	Specification
Diagramming Software	EdrawMax
Operating System	Windows/Linux/macOS
Browser	Chrome/Edge/Firefox, if using online version
RAM	4 GB or above
Storage	2 GB free space
Internet	Required for online/cloud features

4. Objective

The objectives of this experiment are:

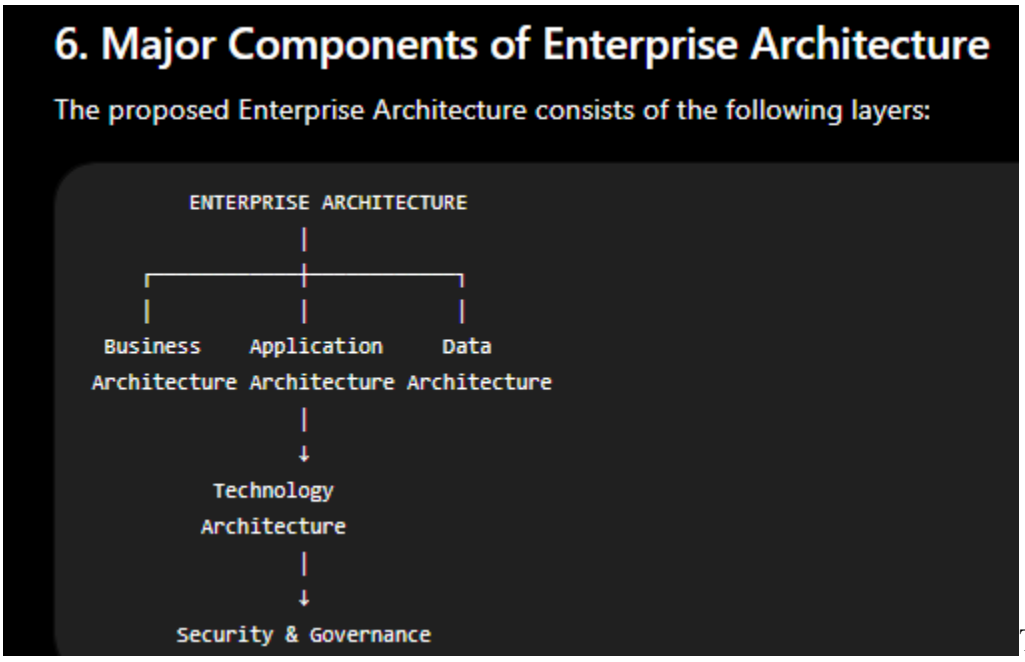
1. To understand the concept of **Enterprise Architecture**.
2. To design an enterprise architecture for a hypothetical organization.
3. To identify the major components of enterprise architecture.
4. To design the **Business Architecture** of the organization.
5. To identify business processes and organizational functions.

5. Theory

5.1 Enterprise Architecture

Enterprise Architecture (EA) is a structured approach used to describe the organization, its business processes, information, applications, and technology infrastructure.
It provides a **holistic view of an organization** and establishes a relationship between business strategy and information technology.
Enterprise Architecture helps an organization to:

- Align IT with business objectives.
- Improve business processes.
- Integrate different applications.
- Manage organizational data.
- Reduce duplication of systems.
- Improve security.
- Support technology planning.
- Facilitate digital transformation.
- Improve decision-making.



The major components are:

1. **Business Architecture**
2. **Application Architecture**
3. **Data Architecture**
4. **Technology Architecture**
5. **Security and Governance**

7. Hypothetical Organization

For this experiment, the hypothetical organization is:

SmartRetail Pvt. Ltd.

Business Domain

SmartRetail Pvt. Ltd. is a retail organization that sells electronic products through **physical stores and an online e-commerce platform**.

The organization manages:

- Customers
- Products
- Orders
- Payments
- Inventory
- Suppliers
- Employees
- Warehouses
- Shipments
- Customer support

The organization wants to integrate its business operations with modern information technology.

8. Business Architecture

Business Architecture describes the **business strategy, organizational structure, business functions, business services, and business processes**.

8.1 Business Vision

"To provide customers with a seamless and efficient digital shopping experience through integrated technology and data-driven business operations."

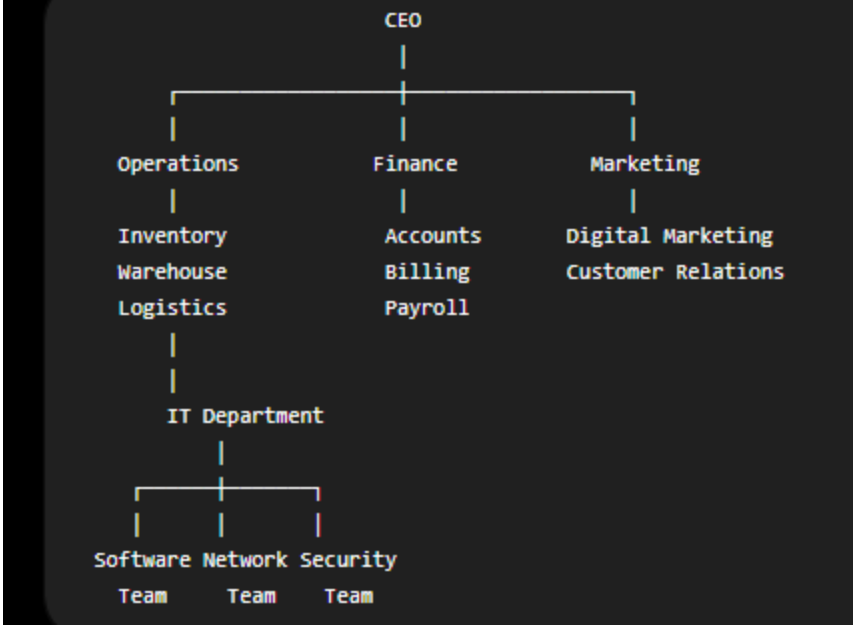
8.2 Business Goals

The major business goals are:

1. Increase online sales.
2. Improve customer satisfaction.
3. Reduce order-processing time.
4. Maintain optimum inventory levels.
5. Improve supply-chain management.
6. Provide faster customer support.
7. Use data analytics for decision-making.

9. Organizational Structure

The proposed organizational structure is:



10. Major Business Processes

The major business processes of SmartRetail are:

Customer Management

Customer Registration
↓
Customer Profile
↓
Product Search
↓
Product Selection

Order Management

Place Order
↓
Order Verification
↓
Payment Processing
↓
Inventory Verification
↓
Order Confirmation
↓
Shipment
↓
Delivery

Inventory Management

Monitor Inventory
↓
Check Stock Level
↓
Reorder Decision
↓
Supplier Order
↓

Goods Received
↓
Inventory Updated

Customer Support

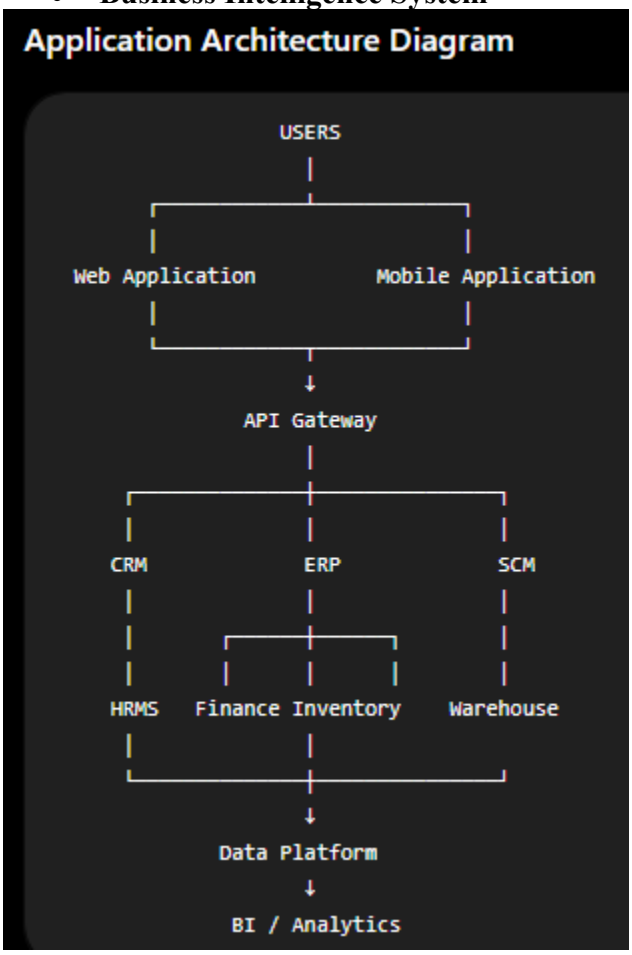
Customer Complaint
↓
Support Ticket
↓
Issue Analysis
↓
Resolution
↓
Customer Feedback

11. Application Architecture

Application Architecture identifies the software applications required to support business processes.

The major applications are:

- **CRM – Customer Relationship Management**
- **ERP – Enterprise Resource Planning**
- **SCM – Supply Chain Management**
- **HRMS – Human Resource Management System**
- **E-Commerce Application**
- **Payment Gateway**
- **Warehouse Management System**
- **Business Intelligence System**

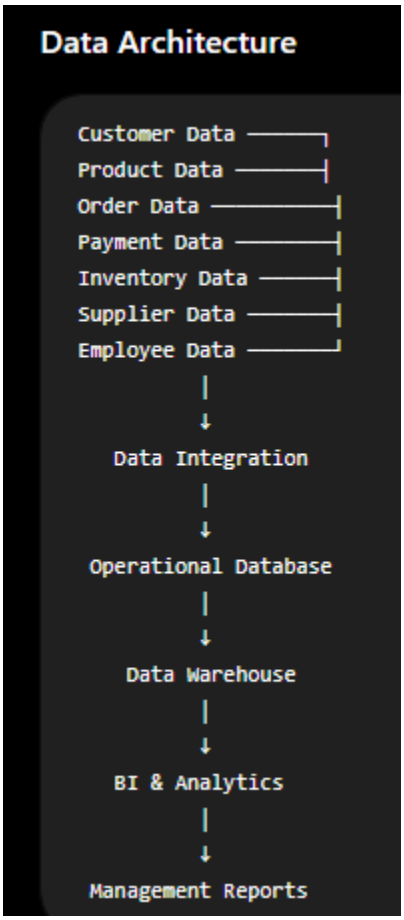


12. Data Architecture

Data Architecture describes the organization's data entities, databases, data flow, storage, and analytics.

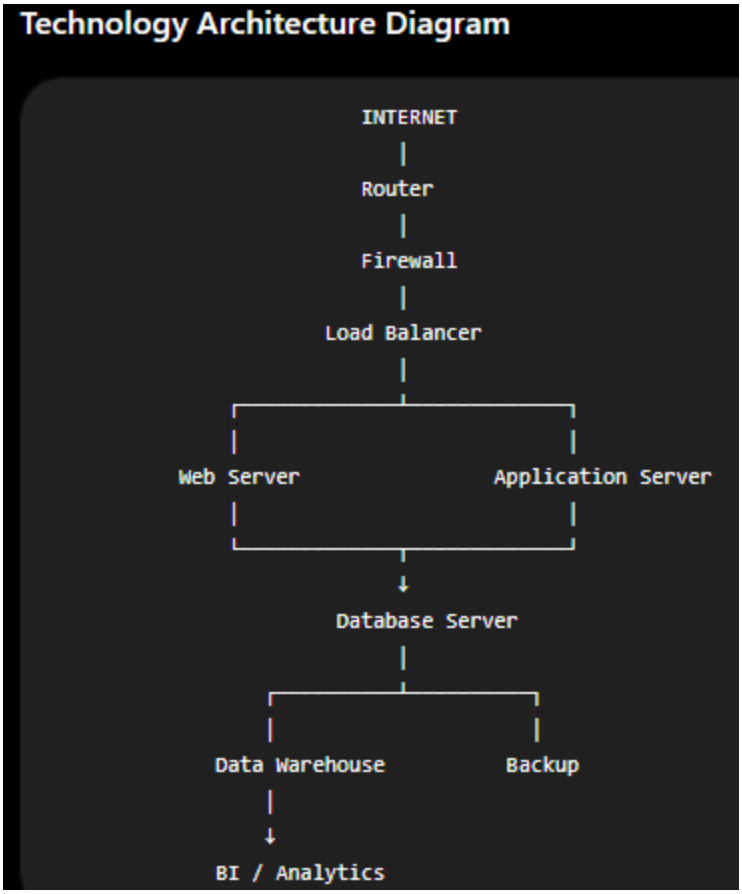
Major Data Entities

- Customer
- Product
- Order
- Payment
- Supplier
- Employee
- Inventory
- Shipment
- Complaint



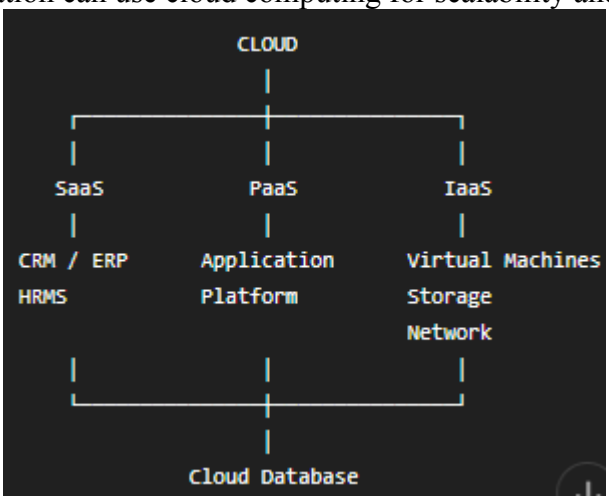
13. Technology Architecture

Technology Architecture describes the hardware, networks, servers, cloud services, storage, and other technical infrastructure required to run applications.



14. Cloud Architecture

The organization can use cloud computing for scalability and availability.



15. Security Architecture

Security is required to protect business applications and organizational data.

Security mechanisms

- Firewall
- User authentication
- Multi-factor authentication
- Role-Based Access Control
- Data encryption
- Secure APIs
- Antivirus/endpoint security

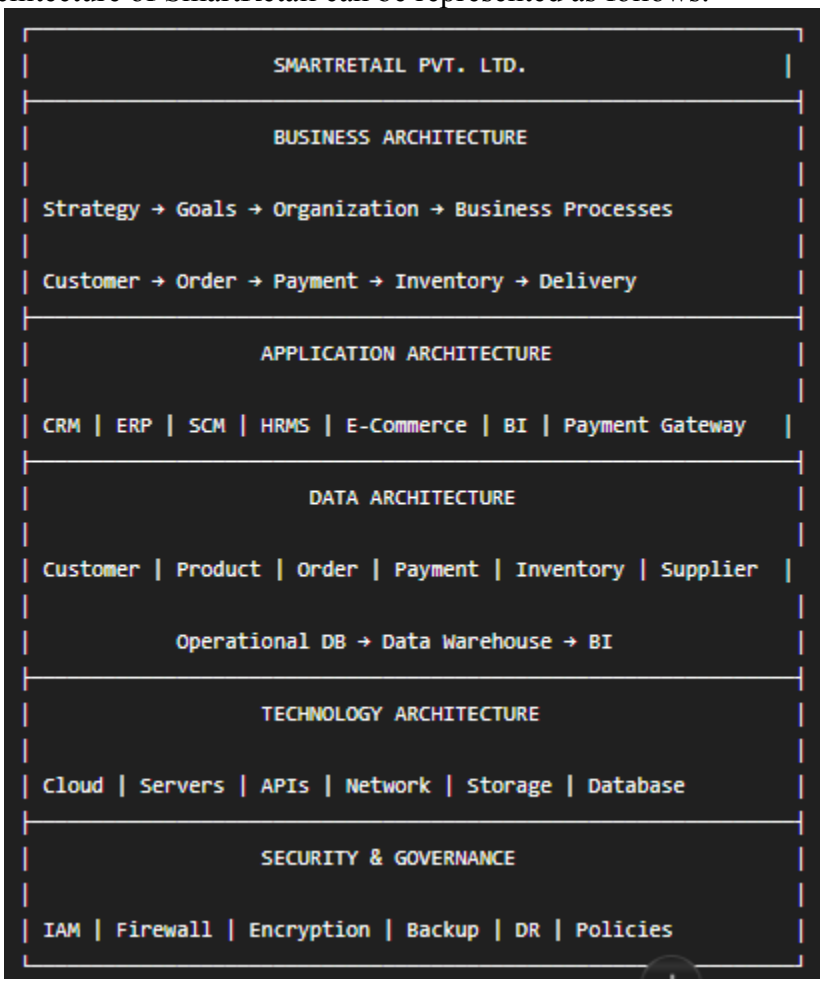
- Backup and recovery
- Network monitoring
- Disaster recovery

Security Flow

User
↓
Authentication
↓
Authorization
↓
Application
↓
Database
↓
Encrypted Data

16. Complete Enterprise Architecture

The complete architecture of SmartRetail can be represented as follows:



17. Implementation Steps in EdrawMax

Step 1: Open EdrawMax

Open **EdrawMax** and select:

New → Enterprise Architecture / Diagram

Step 2: Create Architecture Layers

Create five horizontal layers:

1. Business Architecture
2. Application Architecture
3. Data Architecture
4. Technology Architecture
5. Security & Governance

Step 3: Add Business Components

Add shapes for:

- Strategy
- Business Goals
- Organization
- Business Processes
- Business Services

Step 4: Add Applications

Add:

- CRM
- ERP
- SCM
- HRMS
- E-Commerce
- BI
- Payment Gateway

Step 5: Add Data Components

Add:

- Customer Database
- Product Database
- Order Database
- Inventory Database
- Data Warehouse

Step 6: Add Technology Components

Add:

- Internet
- Router
- Firewall
- Load Balancer
- Web Server
- Application Server
- Database Server
- Cloud
- Backup

Step 7: Connect Components

Use **arrows/connectors** to show dependencies.

For example:

Business Process
↓
Application
↓
Database

↓
Technology Infrastructure

Step 8: Add Labels

Give every component an appropriate label and use different shapes for:

- Business components
- Applications
- Data stores
- Technology infrastructure

Step 9: Save the Architecture

Save the final diagram as:

SmartRetail_Enterprise_Architecture

It can also be exported as **PDF/PNG** for documentation.

18. Architecture Relationship

The main relationship between the architecture layers is:

BUSINESS REQUIREMENTS
↓
BUSINESS ARCHITECTURE
↓
APPLICATION ARCHITECTURE
↓
DATA ARCHITECTURE
↓
TECHNOLOGY ARCHITECTURE
↓
IT INFRASTRUCTURE

For example:

Business Requirement:

Customer should be able to place an online order.

↓

Business Process:

Order Management.

↓

Application:

E-Commerce Application.

↓

Data:

Customer, Product, Order and Payment databases.

↓

Technology:

Web server + Application server + Database server + Cloud infrastructure.

19. Expected Outcome

After completing this experiment, a complete Enterprise Architecture framework for SmartRetail Pvt. Ltd. is obtained.

The architecture provides:

- Clear understanding of business operations.
- Alignment between business and IT.
- Integration of organizational applications.
- Structured management of organizational data.
- Identification of technology infrastructure.
- Better security and governance.

- Improved scalability and maintainability.
- Support for future digital transformation.

20. Advantages

1. **Business-IT Alignment:** Ensures that technology supports business objectives.
2. **Better Integration:** Different applications can communicate with each other.
3. **Improved Decision-Making:** Data can be integrated for analytics.
4. **Scalability:** Cloud and modular applications allow future expansion.
5. **Security:** Security controls can be incorporated at different layers.
6. **Cost Management:** Duplicate applications and infrastructure can be identified.
7. **Improved Performance:** Business processes can be optimized.
8. **Future Planning:** Provides a roadmap for adopting new technologies.

21. Conclusion

The **SmartRetail Enterprise Architecture** integrates business, applications, data, technology, security, and governance. It aligns **business goals with IT systems**, improves integration and scalability, and supports **security and future technology planning** using EdrawMax.